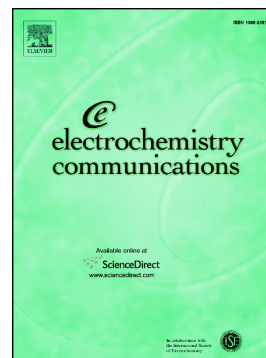


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Ionic Liquid/Boric Ester Binary Electrolytes with Unusually High Lithium Transference Number

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Keywords:

Lithium-ion battery, Binary electrolytes, Lithium transference number, Si-electrode

Abstract:

Binary electrolytes composed of ionic liquids and boric esters were prepared by studying compatibility between various combinations of such systems. The study showed that out of various combinations of ionic liquids/boric esters, only TFSI anion (or FSI anion) based ionic liquids/mesityldimethoxyborane (MDMB) systems were found to be miscible. After equimolar amount of lithium salts was added to ionic liquids, the resulting solution showed high ionic conductivity that was comparable to those for ionic liquids. The lithium transference number (t_{Li^+}) of these systems at room temperature was found to be very high. A maximum t_{Li^+} of 0.93 was observed for a binary mixture of AMImFSI [1-allyl-3-methylimidazolium bis(fluorosulfonyl)imide]/MDMB. Further, this binary mixture as electrolyte in Li/electrolyte/Si cell showed good reversible lithiation-delithiation with more than 2500 mAh/g of delithiation specific capacity.

1. Introduction

Research pertaining to lithium ion batteries (LiBs) have reached its epitome owing to various economic and environmental global scenario [1]. In the field of lithium ion secondary batteries, boron compounds have been utilized in a variety of ways. Boron compounds have been employed as additives with electrolytes, to reduce interfacial resistance between electrode and electrolyte. Moreover, a variety of highly dissociable lithium salts based on lithium borate have been developed. As most well-known example, lithium bis(oxalato)borate (LiBOB) has been widely

employed among researchers of this field today [2-8]. When boron-anion interaction is very strong, marked increase in t_{Li+} is observed instead of decrease in ionic conductivity. A moderate boron-anion interaction does not strongly enhance the t_{Li+} , however, leads to increase of ionic conductivity. [10-15].

As far as we know, boron incorporation to electrolytes was first reported by McBreen *et al* [16]. In the early study, addition of boron compounds in 1,2-dimethoxyethane (DME) solution was found to enhance ionic conductivity. On the other hand, boron incorporation to polymer electrolyte was first reported by Fujinami *et al.* in 1997 [17]. The boroxine ring based network type electrolytes showed significant selectivity for lithium cation transport [17]. This was followed by further molecular design of defined organoboron electrolytes developed by several research groups. For instance, hydroboration polymerization and dehydrocoupling polymerization via hydroborane reagents were found to be useful method for design of defined organoboron polymer electrolytes [10, 13-15].

On the other hand, boron incorporation to molten salt was first reported by Ohno and Matsumi *et al.*, in 2004 [11]. The obtained alkylborane type molten salts showed maximum lithium transference number of around 0.7 due to high Lewis acidity of alkylborane unit. However, this molecular design afforded only solid molten salts showing moderate ionic conductivity of 10^{-5} Scm^{-1} order at 51 °C.

Hence, in the present work, in order to prepare boron incorporated ionic liquid matrices that are liquid at ambient temperature, miscibility of ionic liquids with various boric ester compounds was studied in detail. Some binary electrolytes were found to show high ionic conductivity exceeding 10^{-3} Scm^{-1} due to their lower viscosity than ionic liquid. These electrolytes exhibited unusually high lithium transference number of over 0.9 with electrochemical window of around 5.0 V (Unprecedented properties of the present binary electrolytes are indicated in the graphical abstract). After fabricating a Li/electrolyte/Si half-cell using such a binary electrolyte (AMImFSI/3), reversible charge-discharge with more than 2500 mAh/g of delithiation specific capacity was observed under a moderate coulombic efficiency.

2. Methods

Synthesis of 1-Allyl-3-methylimidazolium FSI (AMImFSI)

To 10.8 g (0.030 mol) of 1-allyl-3-methylimidazolium chloride (AMImCl), aqueous solution of 1.1 eq. of potassium bis(fluorosulfonyl)imide (KFSI) (7.23g, 0.033 mol) was added, and the mixture was stirred for 12h. After the mixture was stirred for another

30 minutes, the aqueous supernatant solution was removed. Then organic phase was washed with distilled water four times, and reprecipitated into diethylether. To the aqueous supernatant solution, silver nitrate test was carried out to confirm the completion of ion-exchange. AMImCl is soluble in water. Hence, unreacted AMImCl in water yields white precipitate with AgNO₃ test. Lack of white precipitation due to AgCl formation evinced completion of the reaction. After removal of supernatant solution, 11.8 g of AMImFSI (yield: 100 %) was obtained after the liquid precipitate was dried under a reduced pressure.

¹H-NMR(CDCl₃): δ (ppm) 4.14 (3 H, -NCH₃),
5.03-5.05 (2 H, -N-CH₂-CH=CH₂), 5.43-5.50 (2H, -CH=CH₂),
5.99-6.09 (1H, -CH₂CH=CH₂), 7.60-7.85 (2 H, -NCH=NCHN), 10.7 (1 H, -NCHN-)

Electrochemical Measurements

Ionic conductivity was measured using a custom designed cell purchased from EC Frontier Co. Ltd. The impedance spectra were recorded on a Solartron 1260 model under frequency range of 1 MHz – 1 Hz. Viscosity of binary solution was evaluated using a corn-plate type viscometer (Bruckfield Co. Ltd.). Lithium transference number of binary electrolytes was measured by the method described by Evans et al [18]. The operations were carried out under an argon atmosphere using identical Li electrodes using potentiostat (Versastat-3; Princeton Applied Research Co. Ltd.). Impedance measurements were carried out under the frequency range of 100 kHz – 10 mHz. A DC potential of 0.03 V (vs Li/Li⁺) was applied during the polarization studies. The linear sweep voltammetry measurements were carried out in a sandwich type cell of the configuration Li/binary mixture electrolyte/Pt. The measurements were performed at the scan rate of 10 mV/sec under the potential range of 0 - 5 V vs Li/Li⁺.

For charge-discharge studies, silicon based anodic half-cells were prepared using CR2025 type coin cells with silicon as the working electrode (diameter = 15 Φ , NISSAN), lithium metal as the counter electrode (15 mm, Honjo metals, Japan) and a polypropylene based membrane (Celgard[®]) as separator (outer diameter = 16 mm, inner diameter= 12 mm). The prepared silicon based anodic half-cells were first discharged and then charged in a galvanostatic mode at 0.3 C-rate with potential cut-offs set between 2.1(OCP) – 0.01 mV (vs Li/Li⁺) using compact charge and discharge system of

EC Frontier; ECAD-1000.

3. Results and Discussion

Firstly, miscibility of various ionic liquids with boric ester compounds was studied (Table 1 and Figure 1a). Under most combinations of ionic liquid/boric ester mixtures, a clear phase separation was observed visually, immediately after mixing. However, only under specific combination of imidazolium TFSI type (or FSI type) ionic liquid and mesityldimethoxyborane, homogeneous solution was readily obtained (Table 1). Moreover, these liquids were miscible under a wide range of volume ratios of ionic liquid and mesityldimethoxyborane. The viscosity of various binary electrolytes under different IL/boric ester volume was measured by corn plate type viscometer, and the results are shown in Figure 1 (b). All the binary mixtures of IL/mesityldimethoxyborane showed significantly lower viscosity in comparison with ionic liquid itself. The viscosity decreased with increase in the volume ratio of mesitylborane. Temperature dependence of viscosity followed Arrhenius behaviour.

These binary mixtures were then subjected to ac-impedance measurements to evaluate their ionic conductivity. This was carried out after adding an equimolar amount of LiTFSI to ionic liquid. Figure 2 shows the temperature dependence of the ionic conductivity for these systems. As expected from low viscosity of the binary solutions, ionic conductivity of these matrices showed small temperature dependency. In spite of the decrease in viscosity, ionic conductivity was found to be similar in range with ionic liquids themselves. This might be possibly due to decreased carrier ion number under efficient anion trapping as discussed below. Ionic conductivity of these binary mixtures was in the range of $1.4 \times 10^{-3} \text{ Scm}^{-1}$ - $9.4 \times 10^{-4} \text{ Scm}^{-1}$ at 51 °C.

The lithium transference number of the binary mixture was measured by combination of ac-impedance and dc-polarization according to the method reported by Vincent, Bruce and Evans et al [18]. All the measurements were carried out in triplicate and was assured that error margin was less than $\pm 1\%$. The changes in lithium ion transference number as a function of concentration of MDMB in three different TFSI⁻ anion based imidazolium ionic liquids are represented in Figure 3(a). All the 1:1 (v/v) binary mixtures showed lithium transference number above 0.5 at ambient temperature. A maximum t_{Li^+} of 0.65 was observed for AMImTFSI/mesityldimethoxyborane binary mixture, which is approximately 3 times higher than that of ionic liquid themselves. The 1:2 (v/v) mixture of BMImTFSI /mesityldimethoxyborane showed even higher lithium transference number of 0.89 with ionic conductivity exceeding 10^{-3} Scm^{-1} . Moreover, lithium ion transference number of AMImFSI/mesityldimethoxyborane reached 0.93 under this volume ratio.

Further increase in boron content (IL/boric ester = 1/3 (v/v)) resulted in the decrease of both ionic conductivity and lithium transference number. This indicated that a highly efficient anion trap retarded ionic conduction due to a fewer number of carrier ion. At the same time, too low viscosity of the system led to unselective migration of both component ions of IL and thereby decreased lithium transference number. To the best of our knowledge, there exists no other comparable organic electrolytes which show more than 0.93 of t_{Li^+} along with ionic conductivity greater than 10^{-3} Scm^{-1} . In the present systems, miscibility of ionic liquids with much amount of boron compound is unique. As a possible reason behind the abnormal observation (high t_{Li^+} values), efficient trap of anion by multiple boron molecules can be assumed.

The electrochemical stability of the binary mixture was studied by linear sweep voltammetry measurements (Figure 3 (b)). The measurements were carried out using Pt/electrolyte/Li cell (vs Li/Li⁺). Although a slight oxidative peak was observed around 2.0 V, moderate electrochemical stability till 4.0 V was observed. AMImFSI/mesityldimethoxyborane system (v/v = 1/2) showed an exceptionally high electrochemical window of around 5.0 V. The oxidative peak that arises at around 2 V is generally attributed to trace amount of water that creeps into ionic liquid system. This reflects in sensitive measurements like cyclic voltammetry. Charge-discharge tests were also carried out after fabricating an anode half-cell composed of Li/electrolyte/Si using AMImFSI/mesityldimethoxyborane (v/v = 1/2) as an electrolyte. In order to focus on inherent capability of electrolytes, simple half-cell set-up was selected for charge-discharge test. The cut-off condition was fixed at 2.1 (OCP) – 0.01 V (vs Li/Li⁺), charge-discharge test was carried out at 0.3 C-rate. As seen in Figure 3(c), reversible lithiation-delithiation was observed with high delithiation specific capacity of more than 2500 mAh/g under a moderate coulombic efficiency (Figure 3(d)). The delithiation specific capacity gradually increased from 2000 mAh/g in 1st cycle to 2500 mAh/g in the 3rd cycle and then was stabilized. This indicates the formation of a robust SEI layer on Si anode. [19]

4. Conclusions

In conclusion, ionic liquid/boric ester binary systems were prepared by simple mixing of mesityldimethoxyborane with imidazolium TFSI (or FSI) based ionic liquids. The obtained binary solution showed significantly lower viscosity in comparison with ionic liquid itself. When lithium salt was added, much improved ionic conductivity in comparison with previously reported organoboron molten salts was observed without significant decrease in lithium transference number. Furthermore, an extraordinarily

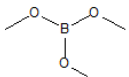
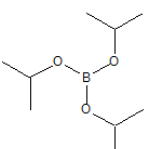
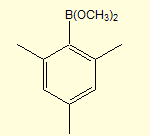
high lithium transference number (0.93) along with high ionic conductivity ($>10^{-3}$ Scm^{-1}) was observed for a 1:2 (v/v) mixture of ionic liquid/boric ester. Further, reversible charge-discharge for Li/electrolyte/Si cell with more than 2500 mAh/g of delithiation specific capacity was observed with moderate coulombic efficiency. This methodology is expected as a new protocol for further improvement of electrolyte systems for improved performance of energy storage devices.

References

- [1] E.M. Erickson, C. Ghanty, D. Aurbach, New horizons for conventional lithium ion battery technology, *J. Phys. Chem. Lett.* 5 (2014) 3313-3324.
- [2] K. Xu, S.S. Zhang, T.R. Jow, W. Xu, C.A. Angell, LiBOB as salt for lithium-ion batteries - A possible solution for high temperature operation. *Electrochem. Solid State Lett.* 5 (2002) A26-A29.
- [3] X.L. Chen, W. Xu, M.H. Engelhard, J.M. Zheng, Y.H. Zhang, F. Ding, J.F. Qian, J.G. Zhang, Mixed salts of LiTFSI and LiBOB for stable LiFePO₄-based batteries at elevated temperatures. *J. Mater. Chem. A*, 2 (2014) 2346-2352.
- [4] J.L. Allen, S.D. Han, P.D. Boyle, W.A. Henderson, Crystal structure and physical properties of lithium difluoro(oxalato)borate (LiDFOB or LiBF(2)Ox). *J. Power Source*, 196 (2011) 9737-9742.
- [5] J. Barthel, M. Wuhr, R. Buestrich, H.J. Gores, New class of electrochemically and thermally stable lithium-salts for lithium battery electrolytes, synthesis and properties of lithium bis[1,2-Benzenediolato(2-)-O,O']borate, *J. Electrochem. Soc.* 142 (1995) 527-2531.
- [6] Z.M. Xue, C.Q. Ji, W.Zhou, C.H. Chen, A new lithium salt with 3-fluoro-1,2-benzenediolato and oxalato complexes of boron for lithium battery electrolytes, *J. Power Sources*, 195 (2010) 3689-3692.
- [7] Z.M. Xue, J.F. Zhao, J. Ding, C.H.Chen, LBD OB, A new lithium salt with benzenediolato and oxalato complexes of boron for lithium battery electrolytes, *J. Power Sources*, 195 (2010) 853-856.
- [8] W. Xu, C.A. Angell, A fusible orthoborate lithium salt with high conductivity in solutions. *Electrochem. Solid State Lett.* 3 (2000) 366-368.
- [9] H.S. Lee, X.Q. Yang, C.L. Xiang, J. McBreen, L.S. Choi, The synthesis of a new family of boron-based anion receptors and the study of their effect on ion pair dissociation and conductivity of lithium salts in nonaqueous solutions. *J Electrochem. Soc.* 145 (1998) 2813-2818.

- [10] N. Matsumi, K. Sugai, H. Ohno, Selective ion transport in organoboron polymer electrolytes bearing a mesitylboron unit, *Macromolecules* 35 (2002) 5731-5733.
- [11] N. Matsumi, M. Miyake, H. Ohno, Molten salts bearing anion receptor. *Chem. Commun.*, 24 (2004) 2852-2853.
- [12] A. Narita, W. Shibayama, K. Sakamoto, T. Mizumo, N. Matsumi, H. Ohno, Lithium ion conduction in an organoborate zwitterion-LiTFSI mixture. *Chem. Commun.*, 18 (2006) 1926-1928.
- [13] N. Matsumi, K. Sugai, M. Miyake, H. Ohno, Polymerized ionic liquids via hydroboration polymerization as single ion conductive polymer electrolytes. *Macromolecules*, 39 (2006) 6924-6927.
- [14] N. Matsumi, K. Sugai, H. Ohno, Ion conductive characteristics of alkylborane type and boric ester type polymer electrolytes derived from mesitylborane *Macromolecules*, 6 (2003) 2321-2326.
- [15] N. Matsumi, T. Mizumo, H. Ohno, Single ion conductive characteristics of poly(organoboron halide)-imidazole complex, *Polym. Bull.* 51(2004) 389-394.
- [16] S. Xu, H.S. Lee, X.Q. Yang, J. McBreen, A new additive for lithium battery electrolytes based on an alkyl borate compound. *J. Electrochem. Soc.* 149 (2002) A355-A359.
- [17] M.A. Mehta, T. Fujinami, Li^+ transference number enhancement in polymer electrolytes by incorporation of anion trapping boroxine rings into the polymer host. *Chem. Lett.* 9 (1997) 915-916.
- [18] J. Evans, C. Vincent, A. P.G. Bruce, Electrochemical measurement of transference numbers in polymer electrolytes. *Polymer*, 28 (1987) 2324-2328.
- [19] S. Dalavi, P. Guduru, B.L. Lucht, Performance enhancing electrolyte additives for lithium ion batteries with silicon anodes. *J. Electrochem. Soc.*, 159(5) (2012) A642-A646.

Table1: Aspect of miscibility under various combinations of ionic liquid/boric esters;
 (-) : not miscible and (+) completely miscible.

IL/Boron		<u>1</u>		<u>2</u>		<u>3</u>
BMImTFSI	-	-	+			
AMImTFSI	-	-	+			
AAImTFSI	-	-	+			
AMImFSI	-	-	+			
EMImFSI	-	-	-			
BMImCl	-	-	-			
AMImCl	-	-	-			
AMImCH ₃ COO ⁻	-	-	-			
AMImBr	-	-	-			

Where, the abbreviations of the cationic part BMIm = 1-butyl-3-methylimidazolium; AMIm = 1-allyl-3-methylimidazolium; AAIm = 1,3-allylimidazolium; EMIm = 1-ethyl-3-methylimidazolium and the anionic part Cl = chloride TFSI = bis(trifluoromethanesulfonyl)imide; FSI = bis(fluorosulfonyl)imide; CH₃COO⁻ = acetyl; Br = bromide

Figure 1(a): Images showing miscibility of representative IL/boric ester (top): completely miscible; (bottom): immiscible. (b): Viscosity profiles showing effect of concentration of boric ester in IL (IL= AMImTFSI; boric ester = mesityldimethoxyborane; V/V)

Figure 2: Arrhenius plots of IL/boric ester mixture (AMImTFSI : mesityldimethoxyborane (MDMB))

Figure 3: Results of electrochemical measurements to validate the IL/boric ester mixture as an electrolyte in LIB. (a): Li transference number (cell; Li/electrolyte/Li); (b): stable potential window measurement (cell; Li/ionic liquid:MDMB (1:2 v/v)/Pt; Vs Li/Li⁺); (c): charge/discharge measurements in anodic half-cell (cell; Li/electrolyte/Si; at 0.3C); (d) columbic efficiency of the anodic half-cell for increasing number of cycle.

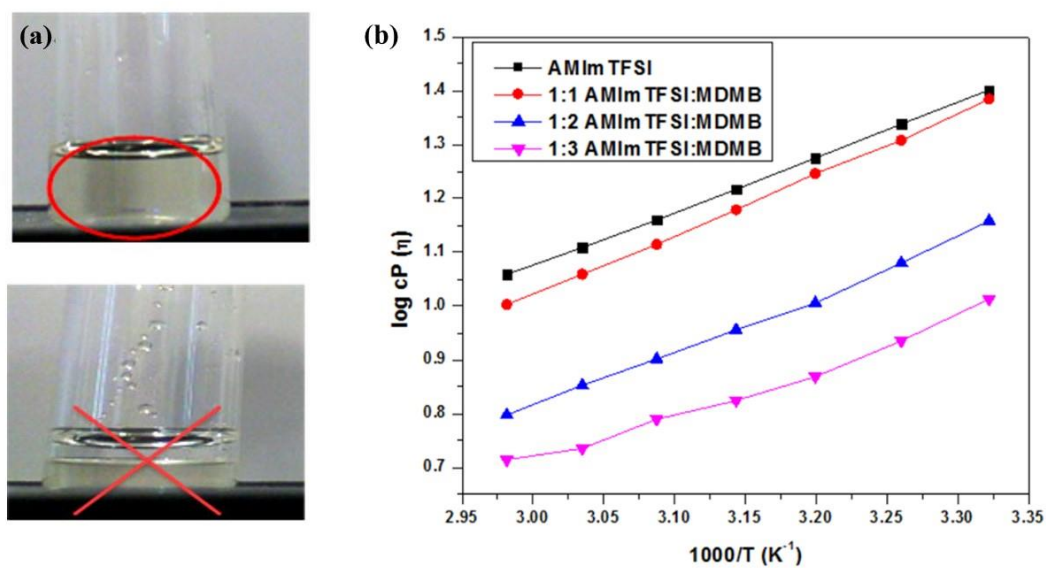


Figure 1

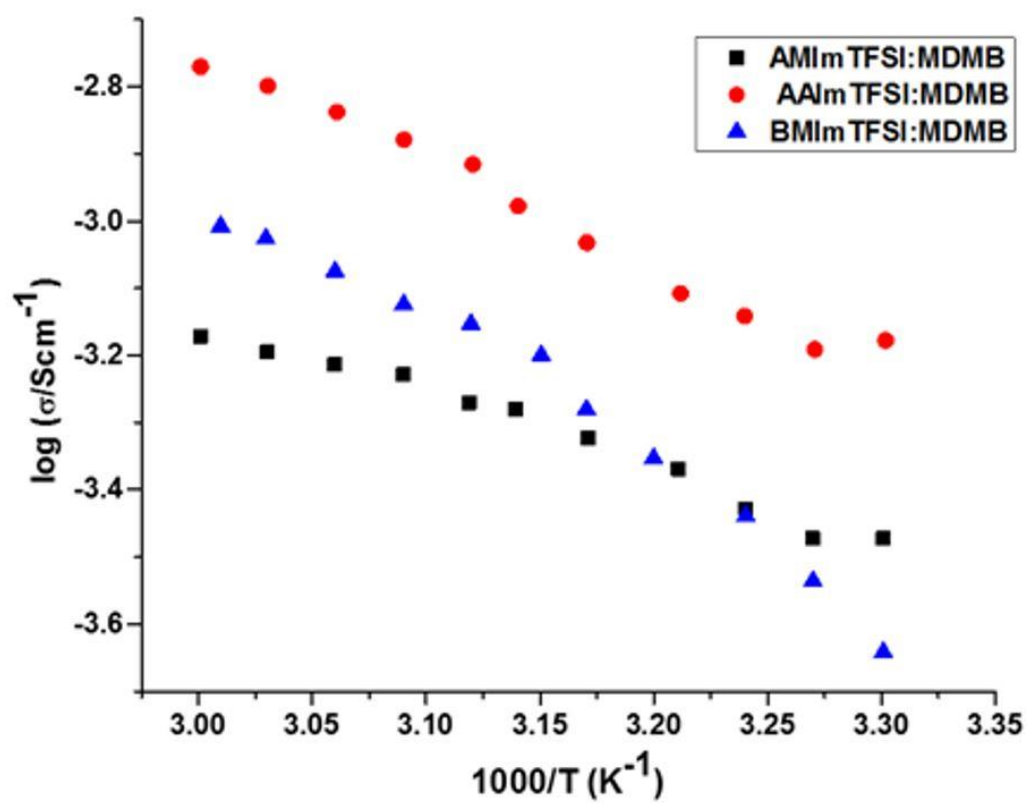


Figure 2

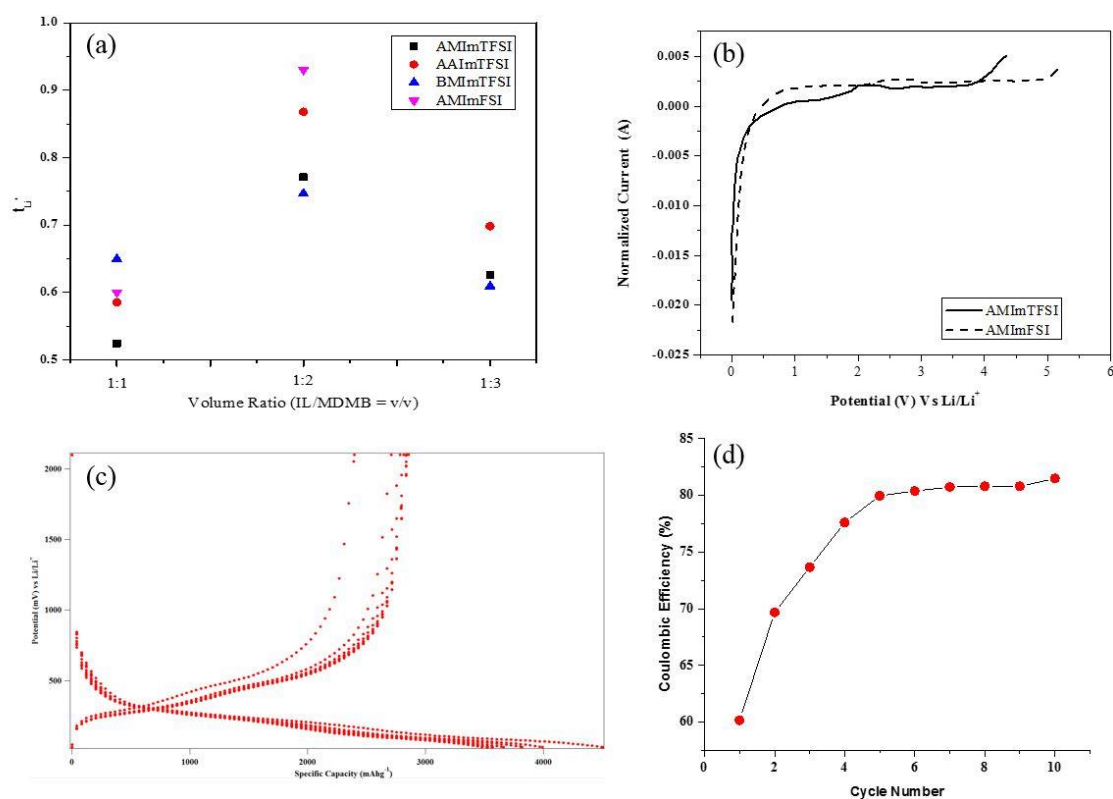
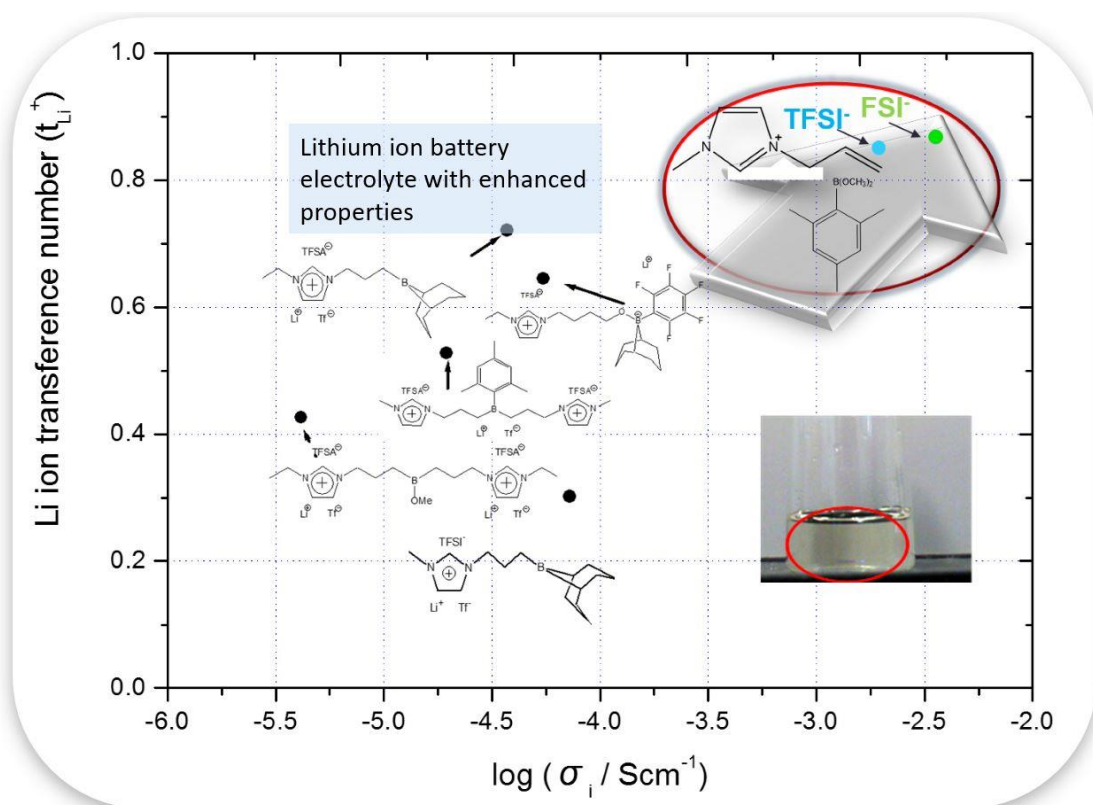


Figure 3



Graphical abstract

Highlights:

1. Several imidazolium TFSI/FSI derivatives were found to be miscible with boric ester.
2. One of the binary electrolytes showed extraordinary high lithium transference number of over 0.9.
3. After fabricating anodic half-cell (Li/electrolyte/Si), specific discharge capacity exceeding 2500 mAhg⁻¹ was observed.